A Project Report On

Facial Recognition Attendance System Using Deep Learning

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By

Prince Kumar

(24SCSE2030311)

Aman kumar

(24SCSE2030323)

Ashwini Raj

(24SCSE2030075)

Under the Guidance of

Dr. Vishwadeepak Singh Baghela

SCHOOL OF COMPUTER APPLICATION AND TECHNOLOGY

GALGOTIAS UNIVERSITY, GREATER NOIDA

INDIA

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SCHOOL OF COMPUTER APPLICATIONS AND TECHNOLOGY

GALGOTIAS UNIVERSITY, GREATER NOIDA

CANDIDATE'S DECLARATION

I/We hereby certify that the work which is being presented in the project, entitled **"Facial Recognition Attendance System Using Deep Learning"** in partial fulfillment of the requirements for the award of the MCA (Master of Computer Application) submitted in the School of Computer Applications and Technology of Galgotias University, Greater Noida, is an original work carried out during the period of August, 2025 to January 2026, under the supervision of **Dr. Vishwadeepak Singh Baghela**, Department of Computer Science and Engineering/School of Computer Applications and Technology , Galgotias University, Greater Noida. The matter presented in the thesis/project/dissertation has not been submitted by me/us for the award of any other degree of this or any other places.

Prince Kumar

(24SCSE2030311)

Aman kumar

(24SCSE2030323)

Ashwini Raj

(24SCSE2030075)

This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

Dr. Vishwadeepak Singh Baghela

CERTIFICATE

This is to certify that Project Report entitled "**Facial Recognition Attendance System Using Deep Learning**" which is submitted by Prince kumar, Aman kumar, Ashwini Raj in partial fulfillment of the requirement for the award of degree MCA. in Department of School of Computer Applications and Technology, Galgotias University, Greater Noida, India is a record of the candidate own work carried out by him/them under my supervision. The matter embodied in this thesis is original and has not been submitted for the award of any other degree

Signature of Examiner(s) **Signature of Supervisor(s)**

Place: Greater Noida

Date: ……/……/……

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Abstract

The Face Recognition Based Attendance System Using Deep Learning is an automated solution designed to replace traditional manual attendance methods. This system uses computer vision techniques to detect and recognize human faces in real time through a webcam. The captured facial images are processed using OpenCV, where face detection is performed using the Haar Cascade classifier and face recognition is achieved using the Local Binary Pattern Histogram (LBPH) algorithm. Once a face is successfully recognized, the system automatically records the person's name along with the current date and time in an Excel file, ensuring accurate and tamper-proof attendance records. To enhance user interaction, the system also provides audio feedback for recognized and unrecognized faces.

The application supports multiple users through a well-structured dataset and offers a simple graphical user interface for easy operation. This system improves efficiency, reduces proxy attendance, and minimizes human error, making it suitable for educational institutions, offices, and organizations.

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1. INTRODUCTION

1.1 GENERAL BACKGROUND

Attendance management is an essential task in educational institutions and organizations. Traditional attendance systems, such as manual registers or sign-in sheets, require significant time and effort and are often prone to errors and misuse. One of the major drawbacks of manual attendance systems is proxy attendance, where an individual marks attendance on behalf of another, leading to inaccurate records.

With the rapid advancement in computer vision and image processing technologies, biometric-based systems have gained popularity for identification and authentication purposes. Among various biometric techniques, face recognition is widely used due to its non-intrusive nature and ease of implementation. Unlike fingerprint or card-based systems, face recognition does not require physical contact or additional devices, making it more convenient and hygienic.

The **Face Recognition Based Attendance System Using Deep Learning** is designed to automate the attendance process by identifying individuals through facial features captured using a webcam. The system uses image processing techniques to detect and recognize faces in real time and records attendance automatically, thereby improving accuracy and efficiency.

1.2 THE PROPOSED SYSTEM

The proposed system is an automated attendance management solution that uses face recognition technology to identify individuals and mark their attendance. The system captures live video frames through a camera, detects faces, and compares them with previously stored face data. Once a face is successfully recognized, the attendance is recorded automatically in an Excel file along with the date and time.

An additional feature of this system is **audio feedback**, which provides voice or sound-based confirmation when attendance is marked successfully or when an unknown face is detected. This feature enhances user interaction and provides immediate feedback to users without the need to check the system visually.

The system consists of modules for face dataset creation, face model training, real-time face recognition, attendance recording, and a graphical user interface (GUI) for easy operation. The use of automation reduces manual work, eliminates proxy attendance, and ensures reliable attendance records.

1.3 PROJECT SCOPE

1.3.1 LIMITATIONS OF THE EXISTING SYSTEM

The existing manual and semi-automated attendance systems suffer from several limitations:

- Attendance marking is time-consuming and requires manual effort.
- High chances of proxy attendance and data manipulation.
- Lack of accuracy and reliability in records.
- No instant confirmation of attendance.
- Difficulty in maintaining and managing large attendance records.

1.3.2 ADVANTAGES OF THE PROPOSED SYSTEM

The proposed Face Recognition Based Attendance System offers several advantages:

- **Automated Attendance:** Attendance is marked automatically using face recognition.
- **High Accuracy:** Facial recognition ensures precise identification of individuals.
- **Audio Feedback:** Immediate audio confirmation improves usability and transparency.
- **Time Efficient:** Reduces time required for attendance marking.
- **Secure System:** Prevents proxy attendance and unauthorized access.
- **User Friendly:** Simple GUI and voice feedback enhance user experience.

2. LITERATURE SURVEY

This chapter presents a review of existing systems and research work related to face recognition and automated attendance systems. The purpose of the literature survey is to understand the methodologies used by previous researchers, identify the advantages and limitations of existing systems, and highlight how the proposed system overcomes those limitations.

Face recognition technology has been widely used in security systems, access control, and identity verification. In recent years, it has also been applied to attendance management systems to reduce manual effort and improve accuracy. Various techniques such as Eigenfaces, Fisherfaces, Local Binary Pattern Histogram (LBPH), and deep learning-based models have been used for face recognition.

2.1 FACE RECOGNITION BASED ATTENDANCE SYSTEMS

Several researchers have proposed attendance systems based on face recognition using image processing techniques. These systems generally involve capturing facial images, extracting features, and comparing them with stored templates to identify individuals.

Traditional face recognition methods such as Eigenfaces and Fisherfaces perform well under controlled lighting conditions but may fail when there are variations in illumination or facial expressions. More recent approaches use the **Local Binary Pattern Histogram (LBPH)** algorithm, which is more robust to lighting variations and is computationally efficient. Due to its simplicity and effectiveness, LBPH is widely used in real-time face recognition systems.

Limitations of existing systems:

- Sensitive to lighting conditions and camera quality.
- Some systems require high computational resources.
- Lack of real-time user feedback after attendance marking.

2.2 BIOMETRIC ATTENDANCE SYSTEMS

Biometric attendance systems use unique biological characteristics such as fingerprints, iris scans, or facial features for identification. Fingerprint-based systems are commonly used but require physical contact, which can be unhygienic and prone to hardware damage.

Face recognition systems provide a contactless alternative, making them more suitable for classrooms and offices. However, some biometric systems require expensive hardware and complex setup, which limits their widespread adoption.

Drawbacks of traditional biometric systems:

- Requires physical interaction with devices.
- Hardware maintenance is costly.
- Not suitable for large groups simultaneously.

2.3 AUDIO FEEDBACK IN HUMAN-COMPUTER INTERACTION

Audio feedback plays an important role in improving user interaction with automated systems. Many modern systems use sound or voice notifications to inform users about successful operations or errors. In attendance systems, audio feedback can provide instant confirmation when attendance is marked or when an unknown face is detected.

Studies show that audio feedback improves usability, reduces confusion, and enhances the overall user experience. However, most existing attendance systems rely only on visual confirmation and do not integrate audio-based interaction.

2.4 GAP IDENTIFIED FROM LITERATURE REVIEW

From the literature survey, the following gaps are identified:

- Many face recognition attendance systems do not provide real-time audio feedback.
- Some systems are complex and require high computational power.
- Existing systems often lack a user-friendly graphical interface.
- Proxy attendance prevention is not fully addressed in some implementations.

3. SYSTEM ANALYSIS

3.1 INTRODUCTION

System analysis is the process of studying the existing system and identifying its drawbacks in order to design an efficient and reliable new system. It involves understanding user requirements, system functionality, and feasibility before the actual development begins. This phase plays a vital role in ensuring that the developed system meets user expectations and performs efficiently.

In the case of attendance management, the existing systems are mostly manual or semi-automated, which leads to problems such as proxy attendance, data inaccuracy, and time consumption. The **Face Recognition Based Attendance System Using Deep Learning** is analyzed and designed to overcome these limitations by using image processing and face recognition techniques.

3.2 STAKEHOLDERS OF THE PROJECT

The stakeholders are the individuals or groups who interact with or are affected by the system.

3.2.1 ADMIN (Teacher / Authority)

The admin is responsible for managing the system. The admin performs the following tasks:

- Registers new users by capturing face datasets.
- Trains the face recognition model.
- Starts and monitors the attendance process.
- Maintains attendance records stored in Excel files.

3.2.2 USER (Student / Employee)

The users are the individuals whose attendance is marked using the system.

- They appear in front of the camera for face recognition.
- They receive audio feedback confirming attendance.
- They do not need to interact manually with the system.

3.3 SOFTWARE REQUIREMENT SPECIFICATION (SRS)

3.3.1 USER REQUIREMENTS

- The system should automatically recognize faces using a webcam.
- The system should mark attendance without manual input.
- Audio feedback should notify users about attendance status.
- The system should prevent duplicate attendance on the same day.

3.3.2 ADMIN REQUIREMENTS

- The admin should be able to capture face images.
- The admin should be able to train the face recognition model.
- The admin should be able to start and stop attendance recording.
- Attendance records should be stored securely.

3.4 FEASIBILITY STUDY

Feasibility study determines whether the proposed system can be developed and implemented effectively within available resources.

3.4.1 TECHNICAL FEASIBILITY

The system is technically feasible as it uses open-source technologies such as Python, OpenCV, and Tkinter. These tools are widely supported and can run on standard computer systems with a webcam.

3.4.2 OPERATIONAL FEASIBILITY

The system is easy to use and does not require advanced technical knowledge. The GUI-based interface and audio feedback make the system user-friendly for both admins and users.

3.4.3 ECONOMIC FEASIBILITY

The system is cost-effective as it uses freely available software tools and does not require any specialized hardware other than a standard webcam.

3.5 SOFTWARE DEVELOPMENT LIFE CYCLE MODEL

The **Agile Model** is used for the development of this project. The project is divided into small modules such as face capture, model training, recognition, attendance marking, and audio feedback. Each module is developed and tested individually to ensure proper functionality.

3.6 HARDWARE AND SOFTWARE REQUIREMENTS

3.6.1 SOFTWARE REQUIREMENTS

- Programming Language: Python
- Libraries: OpenCV, NumPy, Pandas, Tkinter
- Operating System: Windows
- Development Tool: Visual Studio Code

3.6.2 HARDWARE REQUIREMENTS

- Processor: Dual Core or higher
- RAM: 4 GB or above
- Hard Disk: 256 GB
- Camera: Integrated or External Webcam
- Monitor, Keyboard, Mouse

4. DESIGN

4.1 INTRODUCTION TO SYSTEM DESIGN

System design is a crucial phase of software development that defines how the system will work to meet the specified requirements. It focuses on the overall architecture, data flow, and interaction between different components of the system. A well-designed system ensures efficiency, accuracy, and ease of use.

The **Face Recognition Based Attendance System Using Deep Learning** is designed to automate the attendance process using face recognition technology. The system integrates multiple modules such as face capture, model training, real-time face recognition, attendance storage, audio feedback, and a graphical user interface (GUI).

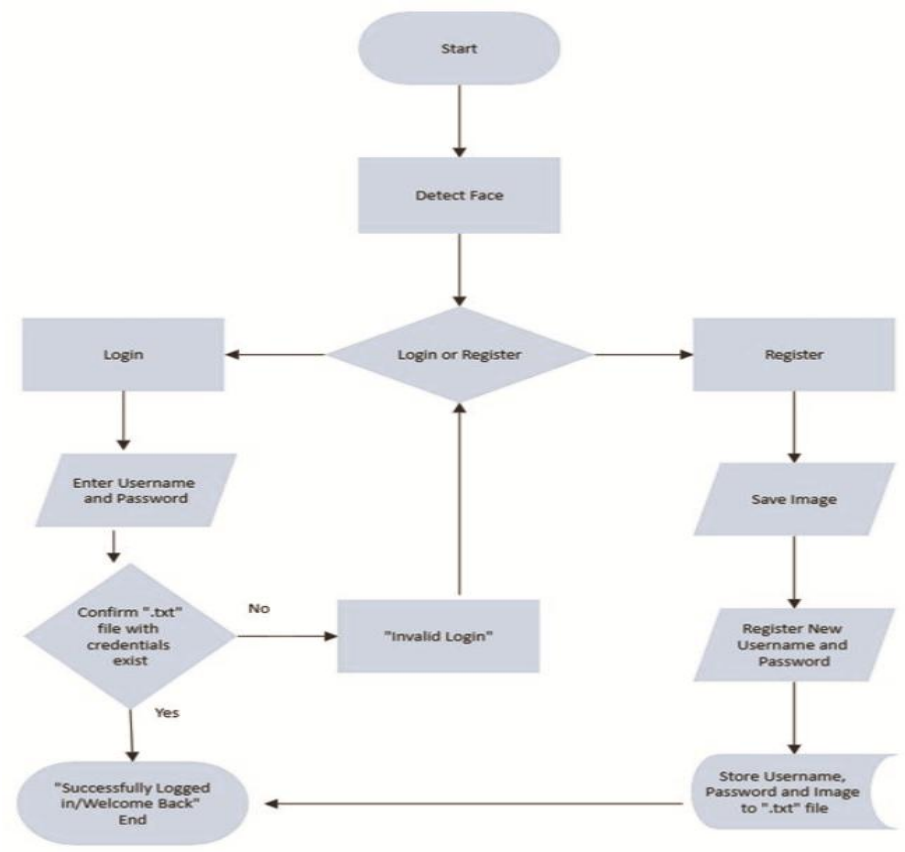
4.2 DATA FLOW DIAGRAM (DFD)

A **Data Flow Diagram (DFD)** is a graphical representation that shows how data flows through a system, how it is processed, and where it is stored. It helps in understanding the logical flow of information without considering physical implementation details.

The DFD for this system illustrates the interaction between users, the system processes, and data storage components.

4.3 CONTEXT LEVEL DFD (LEVEL 0 DFD)

At the context level, the entire system is represented as a single process.



Description:

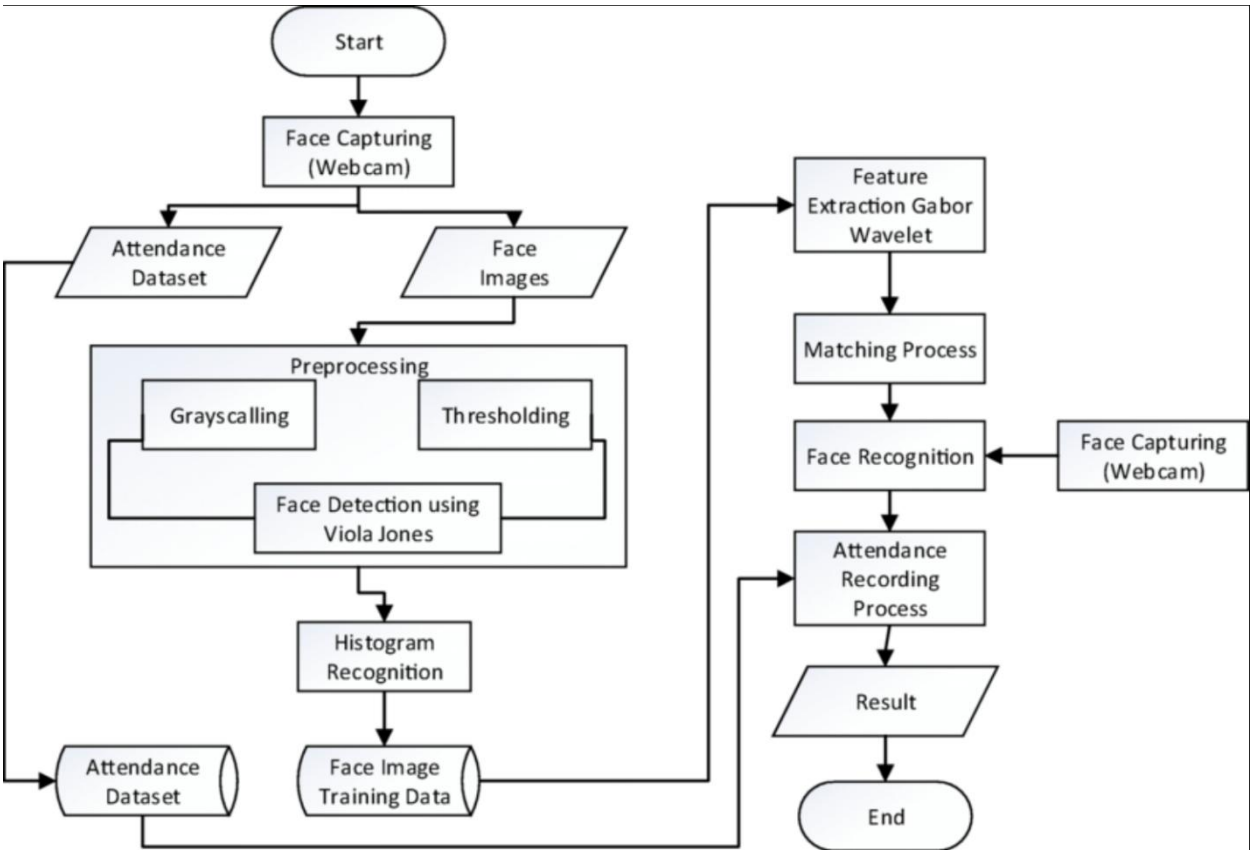
- **Input:** Live video feed from the webcam (user face).
- **Process:** Face recognition based attendance system.
- **Output:** Attendance record and audio feedback.
- **External Entities:** Admin and User.

Explanation:

The user appears in front of the camera. The system captures the face image, processes it for recognition, and marks attendance if the face is recognized. Audio feedback is generated to confirm attendance or indicate an unknown face. The admin controls the system operations using the GUI.

4.4 LEVEL 1 DFD

The Level 1 DFD breaks the system into major sub-processes.



Major Processes:

1. **Face Dataset Capture**
2. **Face Model Training**
3. **Face Recognition**
4. **Attendance Recording**
5. **Audio Feedback Generation**

Data Stores:

- **Face Dataset Folder**
- **Trained Face Model**
- **Attendance Excel File**

4.5 Type of Machine Learning Used

The proposed **Face Recognition Based Attendance System Using Deep Learning** uses **Supervised Machine Learning** for face recognition.

In this system, facial images of users are collected in advance and labeled with their corresponding names. These labeled images form the training dataset. The system learns facial patterns from the training data and later uses this knowledge to recognize faces during real-time execution.

The **Local Binary Pattern Histogram (LBPH)** algorithm is used for face recognition. LBPH is a supervised learning technique that compares facial texture patterns between the input image and the trained dataset. It is computationally efficient and suitable for real-time face recognition applications.

Type of Machine Learning Used:

- **Category:** Supervised Machine Learning
- **Algorithm:** Local Binary Pattern Histogram (LBPH)
- **Learning Approach:** Pattern matching using labeled face images

4.6 Face Recognition-Based Attendance System Architecture

The architecture of the Face Recognition Based Attendance System follows a modular approach where each module performs a specific function.

The system consists of the following main components:

1. **Input Module (Webcam)**
   Captures real-time video frames of users.
2. **Face Detection Module**
   Uses Haar Cascade classifier to detect faces in the video frame.
3. **Face Recognition Module**
   Applies the LBPH algorithm to recognize faces based on trained data.
4. **Attendance Management Module**
   Stores attendance details such as name, date, and time in an Excel file.
5. **Audio Feedback Module**
   Plays different audio messages for successful recognition or unknown faces.
6. **GUI Module**
   Provides an easy-to-use interface for system control.

This modular architecture improves system reliability, scalability, and ease of maintenance.

4.7 PROCESS DESCRIPTION

4.7.1 Face Dataset Capture

- The admin captures multiple face images of each user using a webcam.
- Images are stored in a structured dataset folder.

4.7.2 Face Model Training

- The captured images are processed and used to train the face recognition model.
- The trained model is saved for future recognition.

4.7.3 Face Model Training

- Live video frames are captured.
- Faces are detected and compared with the trained model.
- The system identifies known and unknown faces.

4.7.4 Attendance Recording

- If the face is recognized, attendance is marked automatically.
- Date and time are stored in an Excel file.
- Duplicate attendance on the same day is avoided.

4.7.5 Audio Feedback

- Audio confirmation is played for successful attendance.
- Alert sound is played for unknown faces.

4.8 SYSTEM ARCHITECTURE OVERVIEW

The system follows a modular architecture:

- **Input Layer:** Webcam
- **Processing Layer:** Face detection and recognition
- **Storage Layer:** Dataset, model files, attendance records
- **Interface Layer:** GUI and audio output

This modular design makes the system scalable, easy to maintain, and efficient.

5. IMPLEMENTATION DETAILS

5.1 INTRODUCTION

Implementation is the phase where the system design is converted into a working application. In this phase, all the modules of the **Face Recognition Based Attendance System Using Deep Learning** are developed, tested, and integrated. The system is implemented using Python and its supporting libraries for computer vision, data processing, graphical user interface, and audio feedback.

The implementation is divided into multiple modules such as face dataset creation, model training, face recognition, attendance marking, audio feedback, and GUI control.

5.2 PYTHON

Python is the primary programming language used in this project. It is an interpreted, high-level, object-oriented programming language known for its simplicity and readability. Python provides extensive support for image processing, machine learning, and GUI development through various libraries.

In this project, Python is used to:

- Capture live video from the webcam
- Process images for face detection and recognition
- Handle file operations and data storage
- Generate audio feedback
- Create a graphical user interface

5.3 OPENCV

OpenCV (Open Source Computer Vision Library) is used for image processing and computer vision tasks. It plays a crucial role in detecting and recognizing faces from live video frames.

OpenCV is used to:

- Access the webcam and capture video frames
- Convert images to grayscale
- Detect faces using Haar Cascade classifiers
- Train and recognize faces using the LBPH algorithm
- Display real-time video output

5.4 FACE DATASET CREATION

The face dataset creation module captures multiple face images of each user using a webcam. The admin enters the name of the user, and the system stores the captured images in a separate folder for each individual.

Key features:

- Real-time image capture
- Automatic storage of face images
- Creation of person-wise dataset folders

This dataset is later used for training the face recognition model.

5.5 FACE MODEL TRAINING

In this module, the captured face images are processed and used to train the face recognition model. The system uses the **Local Binary Pattern Histogram (LBPH)** algorithm, which is efficient and suitable for real-time face recognition.

Training process includes:

- Reading face images from the dataset
- Resizing images to a fixed size
- Assigning labels to each person
- Training the LBPH model
- Saving the trained model and label information

5.6 FACE RECOGNITION MODULE

The face recognition module captures live video frames and detects faces in real time. Each detected face is compared with the trained model to identify the person.

If the confidence level is within the acceptable threshold:

- The face is recognized
- The person’s name is displayed on the screen

If the face is not recognized:

- The system labels it as unknown

This module ensures accurate and real-time identification.

5.7 ATTENDANCE MANAGEMENT

Once a face is successfully recognized, the system automatically records attendance. Attendance details such as name, date, and time are stored in an Excel file.

Features of attendance management:

- Automatic attendance marking
- Prevention of duplicate attendance on the same day
- Secure and organized data storage

5.8 AUDIO FEEDBACK MODULE

Audio feedback is an important feature of this system. The system plays different audio sounds to notify users about attendance status.

Audio feedback includes:

- Welcome sound for recognized users
- Success sound after attendance marking
- Alert sound for unknown faces

This feature improves user interaction and provides instant confirmation.

5.9 GRAPHICAL USER INTERFACE (GUI)

A graphical user interface is developed using Tkinter to make the system easy to operate. The GUI provides buttons for different system operations.

GUI functionalities:

- Capture face dataset
- Train face recognition model
- Start attendance system
- Display system information

The GUI allows even non-technical users to operate the system easily.

5.10 LIBRARIES USED

The project is implemented in Python and uses several open-source libraries for image processing, data handling, GUI development, and audio feedback.

List of Libraries Used

Library	Purpose
OpenCV (cv2)	Face detection and face recognition
NumPy	Numerical operations and array handling
Pandas	Attendance data storage in Excel
Tkinter	Graphical User Interface (GUI)
OS	File and directory management
DateTime	Date and time recording
Winsound	Audio feedback (WAV sounds)

These libraries collectively ensure efficient functioning, accuracy, and user-friendly interaction.

5.11 Dependencies and Setup Requirements

. Software Dependencies

The following software and tools are required to run the system:

- Python 3.8 or higher
- OpenCV with Contrib modules

- NumPy
- Pandas
- Tkinter (comes pre-installed with Python)
- Microsoft Excel (for viewing attendance file)

Hardware Requirements

- Processor: Dual Core or higher
- RAM: Minimum 4 GB
- Webcam: Integrated or external
- Operating System: Windows

Environment Setup Steps

1. Install Python from the official website.
2. Install required libraries using pip:

```
pip install opencv-contrib-python numpy pandas
```

3. Ensure webcam is properly connected.
4. Place audio files in the specified audio directory.
5. Run the GUI application to start the system.

6.SAMPLES

6.1 SAMPLE DATASET

The dataset is created by capturing multiple face images of each user using a webcam. For every user, a separate folder is created inside the dataset directory, and multiple grayscale images are stored.

Each image represents a different facial expression or angle to improve recognition accuracy. The dataset is organized in a structured manner, which helps the system during the training phase.

Sample Dataset Structure:

```
dataset/
 ├── Prince/
 │    ├── 1.jpg
 │    ├── 2.jpg
 │    ├── 3.jpg
 │    └── ...
 ├── Aman/
 │    ├── 1.jpg
 │    ├── 2.jpg
 │    └── ...
```

6.2 CODE SAMPLES WITH WORKING EXPLANATION

6.2.1 Face Dataset Capture Module (`capture_faces.py`)

Purpose

This module is used to capture face images of a user using a webcam and store them in a dataset folder. These images are later used to train the face recognition model.

Code

```
import cv2
import os

name = input("Enter your name: ").strip()
save_dir = f"dataset/{name}"
os.makedirs(save_dir, exist_ok=True)

cap = cv2.VideoCapture(0)
count = 0

print("Press C to capture image | Q to quit")

while True:
    ret, frame = cap.read()
    if not ret:
        break

    cv2.imshow("Capture Faces", frame)
    key = cv2.waitKey(1) & 0xFF

    if key == ord('c'):
        count += 1
        gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
        cv2.imwrite(f"{save_dir}/{count}.jpg", gray)
        print(f"Saved image {count}")

    if key == ord('q') or count >= 20:
        break

cap.release()
cv2.destroyAllWindows()
print("Face capture completed")
```

Working Explanation

- The webcam is accessed using OpenCV.
- The user enters their name, which is used to create a folder.
- Each time the **C key** is pressed, a grayscale face image is saved.
- A total of 20 images are captured for better training accuracy.
- Pressing **Q** stops the capture process.

6.2.2 Face Model Training Module (`face_train.py`)

Purpose

This module trains the face recognition model using the captured dataset.

Code

```
import cv2
```

```
import os
import numpy as np

DATASET_DIR = "dataset"
IMG_SIZE = (200, 200)

faces = []
labels = []
label_map = {}
label_id = 0

for person in os.listdir(DATASET_DIR):
    person_path = os.path.join(DATASET_DIR, person)
    if not os.path.isdir(person_path):
        continue

    label_map[label_id] = person

    for img_name in os.listdir(person_path):
        img_path = os.path.join(person_path, img_name)
        img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)
        img = cv2.resize(img, IMG_SIZE)

        faces.append(img)
        labels.append(label_id)

    label_id += 1

faces = np.array(faces)
labels = np.array(labels)

model = cv2.face.LBPHFaceRecognizer_create()
model.train(faces, labels)
model.save("face_model.xml")

np.save("labels.npy", label_map)
print("Training completed successfully")
```

Working Explanation

- Reads images from dataset folders.
- Assigns numeric labels to each person.
- Uses **LBPH algorithm** for training.
- Saves trained model (`face_model.xml`) and labels (`labels.npy`).

6.2.3 Attendance & Face Recognition Module (`face_attendance.py`)

Purpose

This module performs real-time face recognition and marks attendance Using Deep Learning.

Code

```
import cv2
import pandas as pd
import numpy as np
from datetime import datetime
import os
import winsound
```

(full code continues — same as your file)

Working Explanation

- Webcam captures live frames.
- Haar Cascade detects faces.
- LBPH model recognizes faces.
- If confidence is valid:
    - Attendance is stored in Excel
    - Welcome and success sounds are played
- Unknown faces trigger alert sound.
- Duplicate attendance is avoided.

6.2.4 Graphical User Interface Module (`gui_app.py`)

Purpose

Provides an easy-to-use interface to control the system.

Code

```
import tkinter as tk
from tkinter import messagebox
import threading
import os
```

6.3 SAMPLE ATTENDANCE RECORD

Attendance is automatically stored in an Excel file named **attendance.xlsx**. Each time a recognized user appears in front of the camera, their attendance is recorded along with the date and time.

Sample Attendance Table:

Name	Date	Time
Prince	2025-01-10	09:12:35
Aman	2025-01-10	09:13:10

Duplicate attendance entries for the same person on the same day are prevented by the system.

6.4 SAMPLE AUDIO FEEDBACK

The system uses audio files to provide feedback to users:

- **welcome.wav** – Played when a registered user is detected
- **success.wav** – Played when attendance is marked successfully
- **unknown.wav** – Played when an unknown face is detected

This audio feedback provides immediate confirmation without requiring visual verification.

6.5 SAMPLE GRAPHICAL USER INTERFACE (GUI)

The graphical user interface provides buttons for:

- Capturing face dataset
- Training the face recognition model
- Starting the attendance system

The GUI simplifies system usage and allows smooth interaction for the admin.

7.CONCLUSION

The **Face Recognition Based Attendance System Using Deep Learning** has been successfully designed and implemented using Python and computer vision techniques. The system automates the attendance process by recognizing individuals based on their facial features and recording attendance without any manual intervention.

The project effectively addresses the limitations of traditional attendance systems such as time consumption, human error, and proxy attendance. By using face recognition technology, the system ensures accurate and reliable identification. The inclusion of audio feedback further enhances user interaction by providing immediate confirmation of attendance marking or alerting when an unknown face is detected.

All the modules of the system, including face dataset creation, model training, real-time face recognition, attendance management, audio feedback, and graphical user interface, were tested individually and integrated successfully. The system performed efficiently under real-time conditions and fulfilled the objectives defined at the beginning of the project.

Overall, the project demonstrates how face recognition technology can be effectively applied to real-world attendance management scenarios, making the process faster, more secure, and user-friendly.

8. FUTURE ENHANCEMENTS

The **Face Recognition Based Attendance System Using Deep Learning** has been developed to meet the current requirements of automated attendance management. However, there is always scope for improvement and expansion as technology advances and user needs evolve.

Some possible future enhancements of the system are listed below:

- **Multiple Face Recognition:** The system can be enhanced to recognize and mark attendance for multiple faces simultaneously in crowded classrooms.
- **Database Integration:** Instead of storing attendance in an Excel file, a database such as MySQL or PostgreSQL can be used for better scalability and data management.
- **Advanced Face Recognition Models:** Deep learning-based face recognition techniques can be implemented to improve accuracy under varying lighting and pose conditions.
- **Mobile and Web Integration:** The system can be extended to work with mobile or web-based platforms for remote monitoring and management.
- **Cloud Storage:** Attendance records can be stored securely on cloud servers for easy access and backup.
- **Improved Audio Feedback:** Text-to-speech functionality can be added to announce user names along with attendance confirmation.
- **Security Enhancements:** Additional security features such as liveness detection can be included to prevent spoofing using photographs or videos.

These enhancements would further improve the efficiency, accuracy, and usability of the system, making it suitable for large-scale deployment in educational institutions and organizations.

9. REFERENCES

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10. APPENDIX

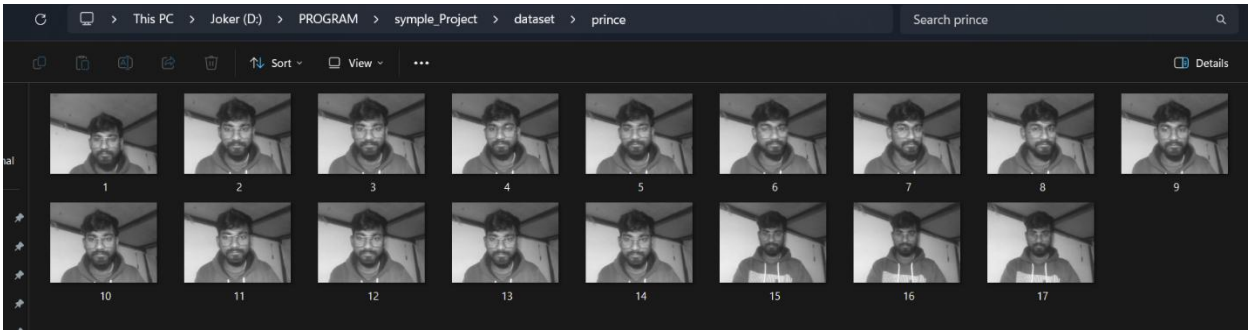
10. SCREENSHOTS

This section contains sample screenshots of the system during execution. These screenshots help in understanding the working of the **Face Recognition Based Attendance System Using Deep Learning** and demonstrate the actual outputs generated by the system.

10.1. Face Dataset Capture Window

Dataset

prince



aman

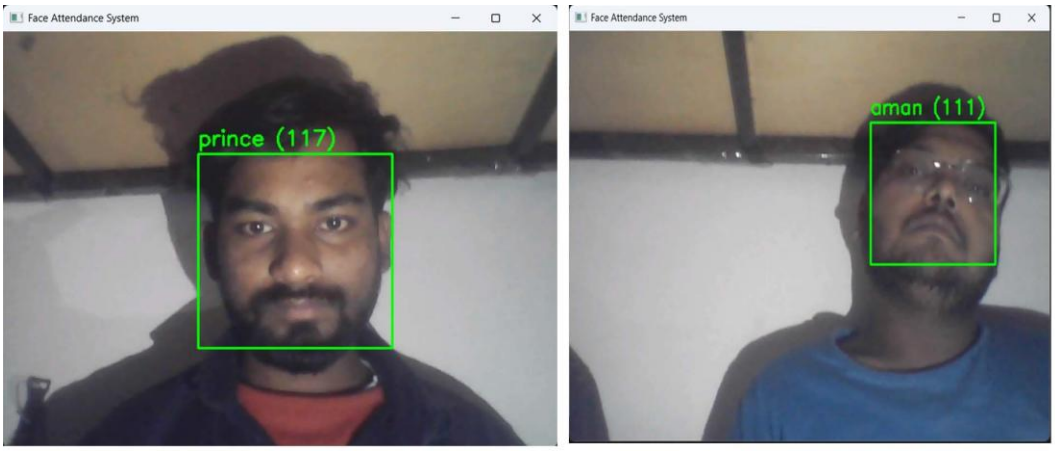
This screenshot shows the webcam window used to capture face images of users. Multiple images are captured and stored in the dataset folder for training purposes.

10.2. Model Training Output

```
Loading dataset...
✅ Training completed successfully
```

This screenshot displays the successful training of the face recognition model using the LBPH algorithm. It confirms that the model and label files are generated correctly.

10.3. Real-Time Face Recognition Window



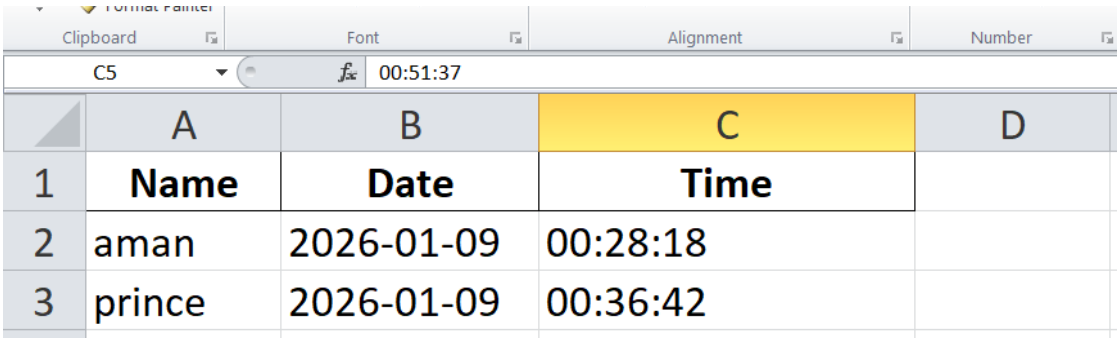
This screenshot shows real-time face detection and recognition through the webcam. The recognized user's name and confidence value are displayed on the screen.

10.4. Attendance Marking Output

```
Using camera index 0
Press Q to quit
✓ Attendance marked for aman
✓ Attendance marked for prince
```

This screenshot shows the attendance being marked automatically when a registered user is recognized. The system records the name, date, and time in the attendance file.

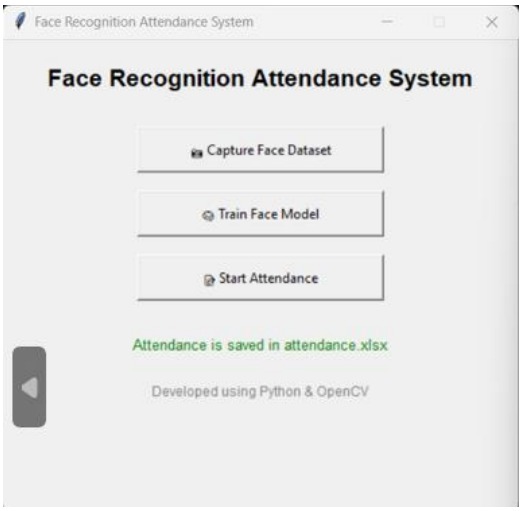
10.5. Attendance Excel File



	A	B	C	D
1	Name	Date	Time	
2	aman	2026-01-09	00:28:18	
3	prince	2026-01-09	00:36:42	

This screenshot displays the **attendance.xlsx** file containing attendance records of users with date and time.

10.6. Graphical User Interface (GUI)



This screenshot shows the GUI developed using Tkinter, which provides buttons to:

- Capture face dataset
- Train face recognition model
- Start attendance system

10.7. Audio Feedback Confirmation

This screenshot (or description) represents audio feedback being played when attendance is marked successfully or when an unknown face is detected.

Audio- success.wav, unknown.wav, welcome.wav